PROJECT 11 BANKING SYSTEM: PYTHON BANK ACCOUNT MANAGEMENT SYSTEM

OBJECTIVE:

Create a Python program that simulates a bank account management system. The system should allow users to create and manage bank accounts, perform transactions, and retrieve relevant details. The project should incorporate the key Python concepts you've learned, such as variables, data structures, functions, file handling, and modules

REQUIREMENTS:

- Account Management:
  - Users should be able to create new bank accounts. Each account must have: Account holder's name, Account number (auto-generated), Account type (e.g., savings or current), and Initial balance.
  - Allow users to retrieve account details, including name, account number, type, and current balance.
- Transactions:
  - Deposit: Users can deposit money into their accounts.
  - Withdrawal: Users can withdraw money, ensuring it does not exceed the current balance.
  - Transfer: Implement a function to transfer funds between two accounts.
  - Ensure transactions update the account balance accordingly.

- File Handling:
  - Store account details and transaction history in files.
  - Implement functionality to read and write account data using the pickle library.
  - Load account data at the start and save updates when the program terminates.
  - Ensure the program can append new transactions without overwriting previous data.
- Reports:
  - Allow users to view transaction history for a specific account, showing date, type, and amount.
  - Generate summary statistics (total deposits, total withdrawals, average transaction amounts) using NumPy functions.
- User Interaction:
  - Create an interactive menu for: Opening an account, viewing details, performing transactions, viewing history, and exiting.
  - Use conditional statements and loops for smooth navigation.
- Error Handling:
  - Implement input validation (e.g., positive deposit/withdrawal amounts).
  - Handle errors like insufficient funds or transfers between non-existent accounts

BONUS (OPTIONAL):

- Implement login functionality for multiple users with usernames and passwords.
- Use advanced Python features like lambda functions for quick calculations